Total No. of Printed Pages: 03

SUBJECT CODE NO: RNP- 8001
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - I
Examination May/June-2026
Applied Mathematics-I

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q. no.1 is compulsory
3) Solve any four questions from Q. nos 2, 3, 4, 5, 6and 7

Q1 Solve any Ten of the following questions **20**

1) By comparison test "If two positive terms $\sum u_n$ & $\sum v_n$ be s. t. $lim_{n\to\infty} \frac{u_n}{v_n} = k$ "then

a) $\sum u_n$ series is convergent or divergent
b) $\sum v_n$ series is convergent or divergent
c) $\sum u_n$ & $\sum v_n$ both series is convergent or divergent
d) None of the above

2) The value of $lim_{x\to 1}\left(\frac{1}{\log x} - \frac{x}{x-1}\right)$ is

a) $\frac{1}{4}$ b) 0 c) -1 d) $-\frac{1}{2}$

3) The series expansion $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$ is of

a) $\cos x$ b) $\tan x$ c) $\sin x$ d) e^x

4) Find the Taylor series expansion of the function $\cos h(x)$ at $x = 0$.

a) $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \infty$
b) $\frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \infty$
c) $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \infty$
d) $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \infty$

5) If u is homogeneous function of x, y of degree n, then by Euler's thm: $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to

a) 0 b) 1 c) nu d) u

6) If $u = log(x^2 + y^2)$ then $\frac{\partial u}{\partial x}$ is

a) $\frac{3x+3y}{x^3y^3}$ b) $\frac{2x}{x^2+y^2}$ c) $\frac{3y}{x^3y^3}$ d) $\frac{3y^2}{x^3+y^3}$

7) Divide 120 into three parts so that the sum of their products taken two at a time shall be maximum if

a) $x = 40, y = 40, z = 40$ b) $x = 30, y = 30, z = 60$

c) $x = 30, y = 45, z = 45$ d) $x = 40, y = 30, z = 50$

8) If u, v & w are functions of three independent variables x, y & z then the Jacobean is given by

a) $J = \frac{\partial(x,y,z)}{\partial(u,v,w)}$ b) $J = \frac{\partial(x,v,z)}{\partial(u,y,w)}$

c) $J = \frac{\partial(u,v,w)}{\partial(x,y,z)}$ d) $J = \frac{\partial(z,y,x)}{\partial(w,y,x)}$

9) The necessary & sufficient condition for D.E. to be exact D.E. is

a) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ b) $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$ c) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$ d) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

10) Newton's second law of motion is

a) $F = m\frac{dv}{dt}$ b) $F = v\frac{dv}{dx}$

c) $F = \frac{dv}{dx}$ d) $F = ma\frac{dv}{dx}$

11) The stationary points of the function $f(x, y) = x^3y^2(1 - x - y)$ are origin

a) $x = 0, y = 1/3$ b) $x = 1/2, y = 0$

c) $x = 1/2, y = 1/3$ d) $x = 1/2, y = -1/3$

12) A circuit containing resistance R and inductance L in series with voltage source E then the D.E for current is

a) $Li + R\frac{di}{dt} = E$ b) $L\frac{di}{dt} + Ri = E$

c) $Li - R\frac{di}{dt} = E$ d) $Li + R\frac{di}{dt} = 0$

13) The Integrating Factor of $\frac{dy}{dx} + xy = x^3$ D.E is

a) $x^2y = c$ b) $x = c$ c) $e^{\frac{x^2}{2}}$ d) $x^2y - xy = c$

Q.2 a) Evaluate: $lim_{x\to 0}\left[\frac{a}{x} - \cot\frac{x}{a}\right]$ **05**

b) Test the convergence of series $\sum\frac{n^2(n+1)^2}{n!}$ **05**

Q.3 a) Solve: $sinx\frac{dy}{dx} + ycosx = 2sin^2x.$ **05**

b) Solve: $cosx\frac{dy}{dx} = -\frac{4x^3y^2+ycosxy}{2x^4y+xcosxy}$ **05**

Q.4 a) If $u = sin^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$ then find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$ **05**

b) If $z = y^x$ then show that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$ **05**

Q.5 a) Expand $\log secx$ **05**

b) Use Taylor's theorem to find $\sqrt{10}$ **05**

Q.6 a) If $xu = vw, yv = wu, zw = uv$ then prove that $\frac{\partial(x,y,z)}{\partial(u,v,w)}$ **05**

b) A rectangular box open at the top is to have volume 256 cubic feet, deter-mine
dimensions of the box required least amount of material. **05**

Q.7 a) A resistance of 100Ω, an inductance of 0.5 Henry is connected in series with a battery of 20 volts. Find the current in the circuit at $t = 0.5$ sec if $i = 0$ at $t = 0$ **05**

b) A body of mass m, falls from rest under gravity in a fluid whose resistance to
motion at any instant is mK times its velocity, where K is a constant. Find the
terminal velocity of the body & also the time to acquire one half of its limit-ing speed. **05**